

Populations of cells in the brain oscillate at various frequencies: delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-12 Hz), beta (12-25 Hz), gamma (25-50 Hz). These are distinct neural rhythms which have been observed consistently throughout the brain and are thought to correspond to different functions, although their exact role is unknown. However, the patterns of these oscillations are very much subjected to environmental influences. Thus, neural oscillations have been shown to synchronise to periodic stimuli, through a process called neural entrainment. During neural entrainment, brainwaves will show greater amplitude in the frequency of the stimulus, and they will show a similar phase to the stimulus in that specific frequency. A classic example of neural entrainment is the auditory steady-state response (ASSR), an evoked potential which arises in response to a series of auditory clicks. EEG data collected from an ASSR experiment will show a peak in amplitude corresponding to the presentation frequency of the stimuli. Moreover, this peak will be greater depending on how often the stimuli are presented, indicating that neural oscillations may have a preferred or optimal frequency of entrainment.

Entrainment is thought to play a critical role in the perception of rhythmic stimuli, and it has been observed to take place in the low frequency range even for irregular stimuli such as speech. While the exact mechanisms of entrainment to speech are not known, one aspect which is relatively well established is that this tracks syllabic rhythm and this has been shown extensively using techniques such as electroencephalography (EEG, refs), magnetoencephalography (MEG, refs) and electrocorticography (ECOG, refs). **Tracking has been investigated as the amount of power or phase locking between neural oscillations in the range of 2-10 Hz, or between the brainwaves and the sound envelope, which carries the slow oscillations in the sound (refs).** There is a great deal of evidence indicating that entrainment is stronger in the delta-theta range than in other frequency ranges (refs), which places the speech envelope at the centre of linguistic processing (refs). However, while the envelope is crucial, it is not the only aspect driving speech perception. In fact, phase locking to slow oscillations is subjected to both bottom up, or stimulus-driven and top down, or higher-level cognitive influences.

As a top down factor, comprehension plays a huge role in the quality and strength of entrainment. Thus, entrainment is lower in the theta range when speech is unattended (refs), unintelligible or noisy (refs), or when participants listen to stimuli in a foreign language (refs). High level properties of speech such as semantic, syntactic and lexical features also affect comprehension, but they do not seem to make a difference to phase locking, especially not in the theta rhythm. Speech-like stimuli can still trigger phase locking between 4 and 7 Hz, even when intelligibility is affected. For example, Howard and Poeppel (2010) showed that phase coherence was intact when they played speech stimuli backwards, in the absence of any distinguishable linguistic structures. However, there is evidence of entrainment to syntactic units in speech, such as one study (Ding and Simon,) which showed that distinct units (monosyllabic words, as well as the phrases and sentences which they are part of) elicit peaks in power at the frequency with which they are presented, indicating a clear influence of linguistic processing.

On what concerns purely acoustic information, research suggests that there may be two temporal windows which are activated during sound processing, and possibly during entrainment to speech in particular. Using noise stimuli which were frequency-modulated at 4, 10 and 50 Hz, Luo and Poeppel (2012) showed in their MEG study that phase locking in the auditory channels happens only in the theta and gamma bands. Thus, it has been suggested that while the theta rhythm tracks the syllabic tempo, gamma waves respond to fine structure and phonetic information (Poeppel, 2003). Psychophysical evidence supports the idea that fine structure is also important to entrainment. Zoefel and VanRullen (2014) found that when envelope fluctuations are filled with noise, but fine

structure is left intact, participants are still able to detect clicks inserted at the onsets of syllables. Moreover, it seems that speech-like sounds with their fine structure still intact but no amplitude modulations are still able to elicit entrainment in the theta range. Nevertheless, speech is reported more unintelligible when the low frequency modulations are removed from the sounds, rather than when the fine structure is altered (Doelling et al, 2014).

Phonetic and syllabic information may influence each other through theta-gamma coupling (Giraud and Poeppel, 2012), but while theta seems to play the main part in speech perception, the exact direction of this coupling remains unclear. Furthermore, like briefly mentioned above, it seems to be the syllable onsets which play a critical role during entrainment. Doelling et al. (2014) showed that when only envelope information is maintained within the stimuli, and clicks are inserted either alongside envelope points which corresponded to either the beginning or the middle of the syllable, both intelligibility and entrainment in the theta range were higher in the first scenario. However, theta remained higher than gamma, and entrainment in the latter did not differ between conditions.

Subjects

Subjects were 21 adults (14 females, mean age, sd) who were paid £10/hour to participate in the experiment. All participants were native English speakers with minimal knowledge of Russian (for example, we decided it to be fine to participate if one had studied Russian for a length of two years maximum, more than 10 years prior to the experiment, and had not practiced it in the meantime). Eligibility requirements were normal hearing, right-handed and no known history of neuropsychological conditions with an emphasis on learning disabilities.

Design

We wanted to investigate the effects of edges in natural language and whether these are influenced by intelligibility. We manipulated intelligibility by having two language conditions, one native (English) and one unknown to the participants (Russian). The study followed a 2 (language) x 3 (edge) factorial design. In addition to the strong and weak edge conditions, we had mixed or filler stimuli (syllable onsets were defined by a variety of sounds, see below), which acted as a control condition.

Stimuli

Stimuli were 100 sentences spoken by a bilingual English-Russian speaker and recorded in a soundproof room using Equipment. For each language, we used 20 strong sentences which we paired with weak sentences in the sense that they had the same number of syllables, syntactic structure, intonation and stress. We also added a similar filler sentence to 10 of these pairs. Each sentence was repeated four times. Examples of the stimuli can be seen below:

English Peg-gy dic-ta-ted pa-pers to ty-pists.	strong
La-rry fal-si-fied lea-ses for fe-lons.	weak

	Ba-rry im-por-ted chi-cken from trades-men.	filler
Russian	Ba-ba ka-tit ba-gazh po tro-pin-ke.	strong
	<i>The woman is strolling luggage along the path.</i>	
	Ljo-va lo-vit vo-ron za za-li-vom.	weak
	<i>Leva is catching crows behind the bay.</i>	
	Pe-tja vy-nul lis-ty iz kar-ma-na.	filler
	<i>Petja has taken out papers from the pocket.</i>	

We included a cough in one of the filler presentations, which always appeared 1.6 seconds after the stimulus onset and lasted 400 ms. In total there were 410 stimuli. Those which included a cough were thus shorter than the shortest sentence without a cough (2.1 seconds). Average stimulus duration was 2.3 seconds (x for English and y for Russian, w for strong, z for weak, u for filler, no significant difference between the groups), and the number of syllables in each sentence varied between 8 and 11. We also computed the mean syllable duration for English and Russian sentences, which was 230 ms for each group, and not that far from the median of each group (220 ms). Importantly, our syllable length distribution shows a gamma distribution like it normally does for all natural languages, despite the median being lower than normal in each condition (add figures). All stimuli were normalised to 70 dB loudness.

Apparatus

We used 32-channel Brain Products EEG to conduct scalp activity. The setup required inserting Electrolyte gel through the electrode holes and onto the scalp to increase conductivity. The stimuli were delivered using Presentation software and through a pair of ... headphones placed comfortably onto the EEG cap. Participants adjusted the volume at the level they saw fit. They were asked to fixate onto a cross on the screen in front of them (model: ...) in order to avoid closing their eyes (and more EEG power in the alpha frequency range). EEG activity was recorded at 1KHz sampling rate using actiCap equipment and Vision Recorder software (BrainProducts Ltd.).

Procedure

The experiment lasted approximately 1.5 hours including setup (i.e., gelling session, 40 minutes). The experimental session was 35 minutes long. Participants were asked to fixate onto the cross as much as possible, and not move or blink during the delivery of the stimulus. The only task was to press the 'Space' bar as soon as their heard a cough. The presentation of the stimuli was divided into 5 trials each containing 82 stimuli. There was a 30 second break between each trial, but participants could take longer if they wished, in order to minimise fatigue.

Data Analysis

All data analyses (for both the stimuli and EEG data) were conducted in Matlab R2016b. Pre-processing of EEG data was done using EEGLAB toolbox for Matlab. Data were low-pass filtered at 50 Hz, re-referenced using average reference, and then visually inspected for paroxistic artifacts. ICA decomposition was used to detect blinks, eye movements and ECG activity. Further time-frequency analyses were done using custom Matlab scripts. First, we took the spectrogram of the EEG signal and that of each sound file, which we computed in 10 ms steps for frequencies between 1 and 40 Hz, with a frequency resolution of .5 Hz below 10 Hz and 1 Hz above. We used the

spectrograms in our calculations for the phase coherence between the EEG signal and that of the sound envelope, as well as that between individual trials, using equation (1):

$$C_{phase_{ij}} = \left(\frac{\sum \cos(\theta_{ij})}{N} \right)^2 + \left(\frac{\sum \sin(\theta_{ij})}{N} \right)^2 \quad (1),$$

where θ_{ij} is the phase difference between two trials, or between the sound and the EEG, and N is the number of time points in a trial. We used the sound envelope and not the entire sound because we were mainly interested in changes that occur at the syllabic rhythm, which is mainly conveyed by the envelope. We measured the power covariance between different EEG trials only, like in the equation below:

$$C_{power_{ij}} = \frac{\sum_N (|Pow_i| - \mu_i)(|Pow_j| - \mu_j)}{\sigma_i - \sigma_j} \quad (2),$$

Where i, j are different trials, $|Pow_i|$ is the absolute power at trial i, μ_i is its mean and σ_i is its standard deviation. We computed the phase coherence between EEG trials and their corresponding sound envelopes, as well as to random sounds, the latter of which quantified the noise in our measure. We then subtracted the coherence to random sounds from that to actual sounds, which then gave a measure of de-noised phase coherence. Similarly, we computed the power coherence between all the possible pairs of our 4 trials to the same stimulus, names intra-trial coherence, then extracted the power coherence to pairs of randomly sampled trials. Like Luo and Poeppel, we named this the phase dissimilarity function which can be seen in equation (5). This gave us a measure of de-noised phase coherence, and we repeated the process for power covariance, like in equation (6).

$$Phase - dissimilarity = \frac{\sum_{j=1}^J C_{phase_{intra}}}{J} - \frac{\sum_{j=1}^J C_{phase_{inter}}}{J} \quad (5)$$

$$Power - dissimilarity = \frac{\sum_{j=1}^J C_{power_{intra}}}{J} - \frac{\sum_{j=1}^J C_{power_{inter}}}{J} \quad (6)$$

Where J is the number of trials.

*** mention zscores

Stimulus sharpness was calculated by taking the envelope of each sound file, using the analytic signal of the Hilbert transform. The envelope's first derivative was then normalised to the envelope's area to ignore raises in envelope produced by isolated sounds such as terminal 's' consonants which could unnecessarily add to the sound's sharpness measure.

Results

Stimuli

Statistical analyses were conducted on various properties of our stimuli to understand in which way these might have affected EEG data. The sentences in each condition were of very similar durations

(overall mean was 2.45 seconds, lowest mean was 2.36 seconds – English strong - and highest mean was 2.51 seconds – Russian weak) and there was no difference between conditions. Similarly, there was no difference between the mean (approx. 230 ms) and median (approx. 220 ms) duration of the syllables in each condition.

It is a well known fact that syllable durations in each language follow a gamma distribution, and their median is known for each language. We also computed histograms of our syllable durations, and these had the shape of a gamma distribution in both English and Russian, as shown in Figures 1 and 2.

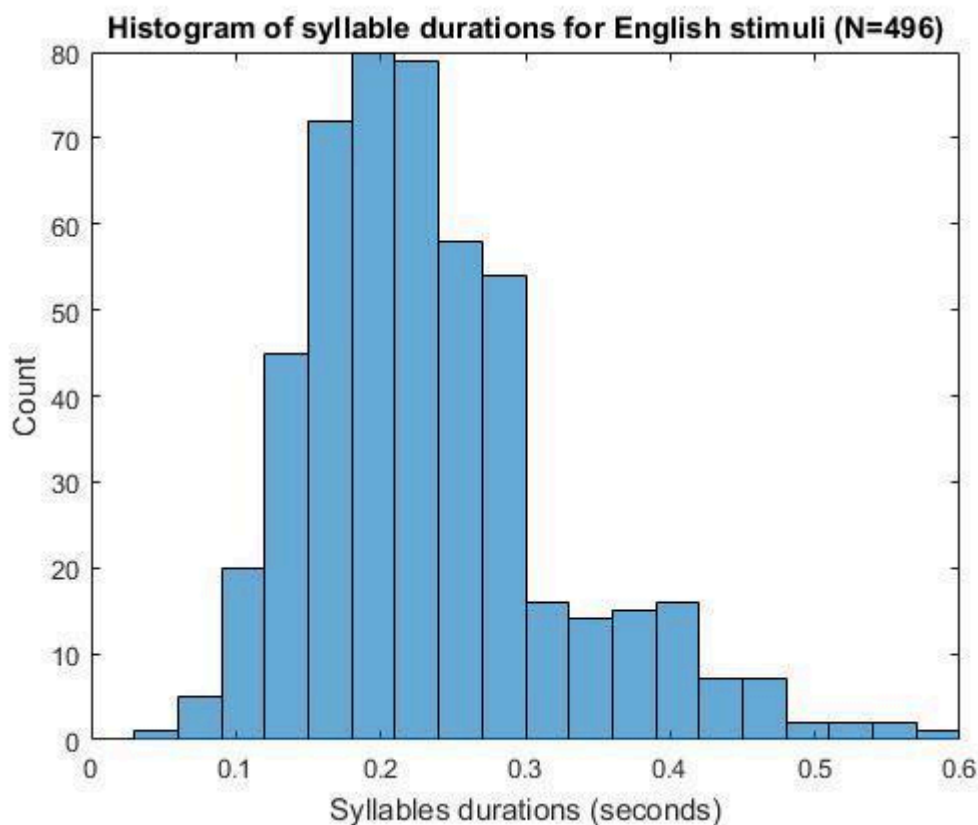


Figure 1. Histogram of syllable durations in all English sentences. The number of syllables is plotted as a function of syllable times measured in seconds.

Sharpness values indeed showed that strong stimuli altogether were sharper than weak stimuli ($t(78)=8.32$, $p<.0001$), but there was no difference in sharpness between English and Russian stimuli. Strong stimuli were sharper in each language condition (English, $t(38) = 5.51$, $p<.0001$, Russian, $t(38) = 6.46$, $p<.0001$). These results can be clearly seen in Figure 3.

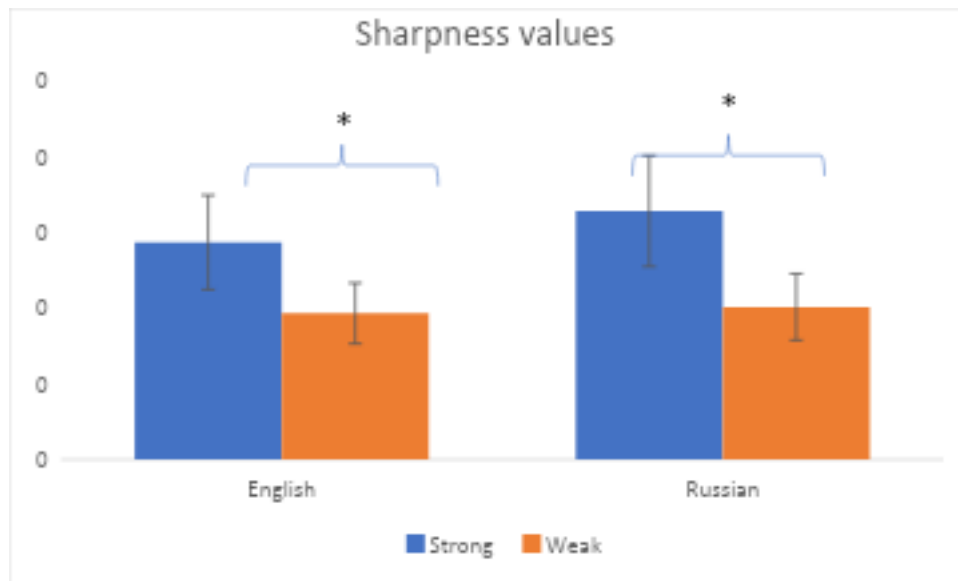


Figure 3. Sharpness values are plotted as a function of condition. Stars indicate significant differences between conditions.

Figure 4 shows an example of a pair of weak and strong sentences in English. Because weak stimuli seemed to have higher overall mean amplitude, we also computed the overall amplitude of each sound file using the root mean square of the sound envelopes. Indeed, weak stimuli had significantly higher RMS values than strong stimuli ($t(78) = -2.46$, $p < .01$), but so did Russian sounds compared to English ones ($t(78) = -2.76$, $p < .01$).

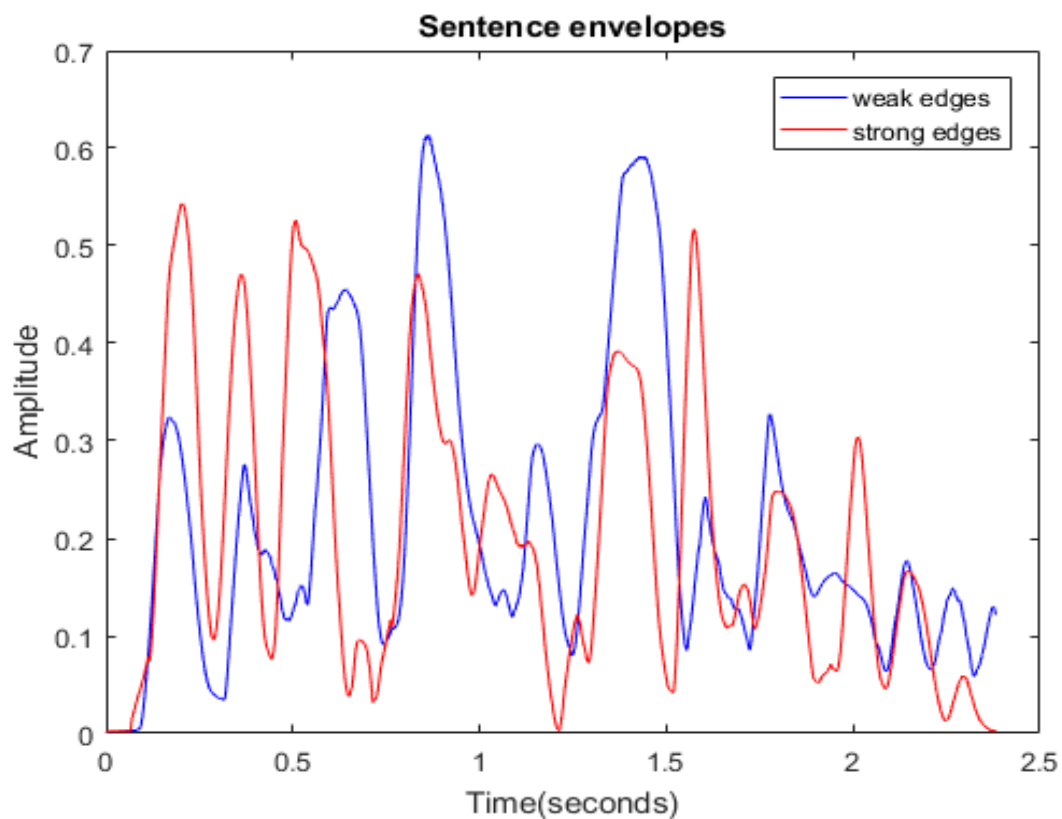


Figure 4. Sound envelopes of a pair of weak and strong stimuli. The amplitude of the envelope is plotted as a function of time.

To test whether the theta rhythm would be detectable, we computed the power spectrum of the sound files. We did this for frequencies between 1 and 40 Hz, with a frequency resolution of 0.5 Hz for frequencies between 1 and 10 Hz, and that of 1 Hz for frequencies between 10 and 40 Hz. The power spectrum of each condition can be seen in Figure 5. There is a clear peak at 5 Hz in the strong conditions, for both the English and Russian languages. The highest power in the weak conditions, on the other hand, seems to be smeared between 2.5 and 4 Hz. Power, thus, seems to be higher for Russian than English sounds ($t(39)=-4.16$, $p<.0001$) and for Strong than Weak sound envelopes ($t(39)=7.66$, $p<.0001$). Thus, if intelligibility played no part in neural entrainment, one would expect this to be greater in the Russian condition, but also when listening to stimuli with strong acoustic edges.

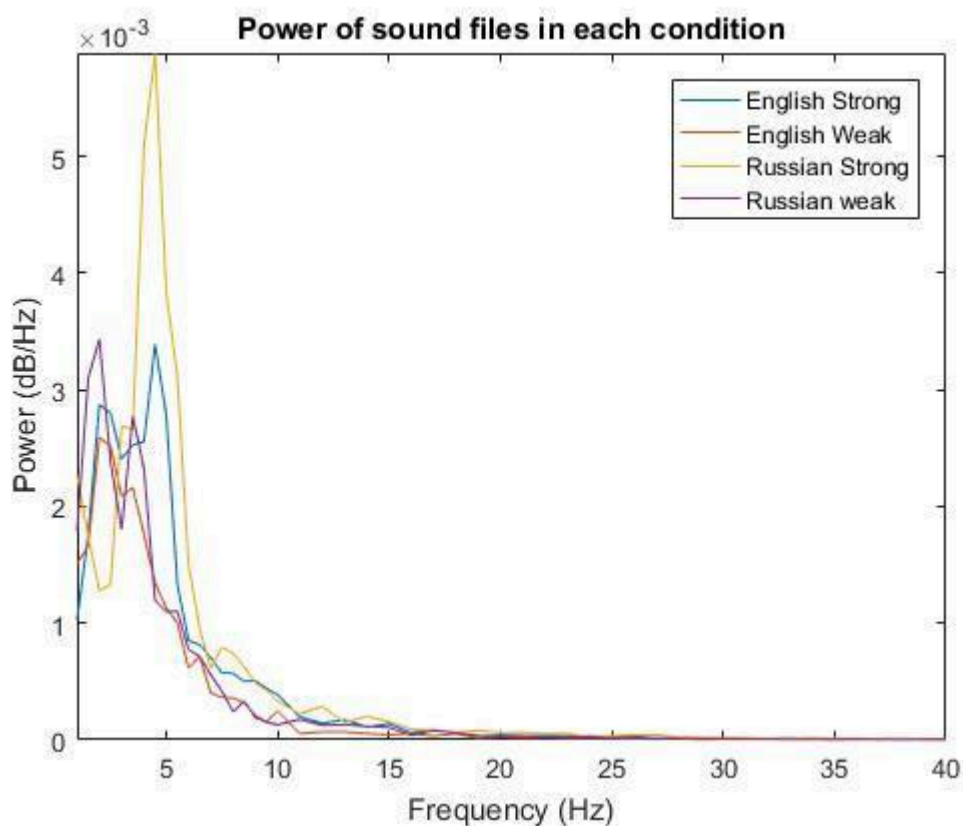


Figure 5. Average power (dB/Hz) of stimuli in each condition is plotted as a function of frequency (Hz).

We left the filler stimuli out of the analysis due to the smaller number of sentences in this group. However, EEG data in response to filler stimuli, as well as to the rest of the sentences, were used for purposes of channel selection when computing phase coherence and power covariance, the two main quantities of interest in our analysis of entrainment.

EEG

To determine channel selection for future data analyses, we first plotted heatmaps depicting the power spectra in each frequency range across the scalp. As one can see, theta seems to be localised at midline and occipital electrodes, alpha only at occipital sites and gamma is lateralised at frontal

electrodes. Alpha generally tends to localise at occipital sites, but theta does not, whereas gamma scalp distribution is greatly influenced by the task (references needed). Due to the auditory nature of our experiment, the well-known distribution of the N1 ERP at midline electrodes and because we did see considerable activity in the theta band at central sites, we decided to conduct all further analyses on such channels, mainly Fz, Fc1, Fc2 and Cz. Plots and statistical analyses of EEG data were then conducted by taking the average of these electrodes (consider revising since the average wasn't done until the end of the analyses).

In the heatmaps, we looked at specific points within the frequency bands to be able to localise specific topographical subsets within each range. However, we did not seem to notice distinct clusters unless those obtained across frequencies. Therefore, when comparing differences between bands, we took the average between specific frequency points, defined as follows: for theta, these were between 3 and 7 Hz, alpha, 8 and 13 Hz, gamma, 25 and 40 Hz.

Power analyses entailed the highest peak in the alpha range, with or without $1/f$ noise, as shown in Figures 6 and 7. There was however, no difference between conditions in either frequency bands. Because we detected no activity in the gamma band when $1/f$ noise was present in the data, we only report statistical tests referring to the second case. Thus, a one-way ANOVA between frequency bands was significant ($F=239.54$, $MSE=0.02463$, $p<.0001$), with greatest power being shown in the alpha range, which was significantly greater than both theta and gamma, as shown by post-hoc T-tests (Theta vs alpha, $t(20)=-5.55$, $p<.0001$, Alpha vs Gamma, $t(20)=3.51$, $p=.002$). Theta was also significantly lower than gamma ($t(20)=-11.47$, $p<.0001$), and there was a slight effect of language in both the alpha and gamma ranges. Alpha was higher in the English than in the Russian condition ($t(20)=2.21$, $p=.019$), and the opposite was found for gamma ($t(20)=-2.79$, $p=.005$). While findings in the gamma range can potentially be influenced by properties of the stimuli, we cannot draw a conclusion since these are not the sites where gamma has the most power. The findings in the alpha range are, nonetheless, surprising, since we did not expect a difference between conditions in this frequency interval. Concerning is also the fact that the peak in the theta range is small and quite late (7 Hz), and we only obtain a marginal effect of language in the expected direction, namely that theta is higher in the English condition than in the Russian one. One thing worth noting is that another significant peak seems to be in the beta range, but that expands outside the purpose of our analysis.

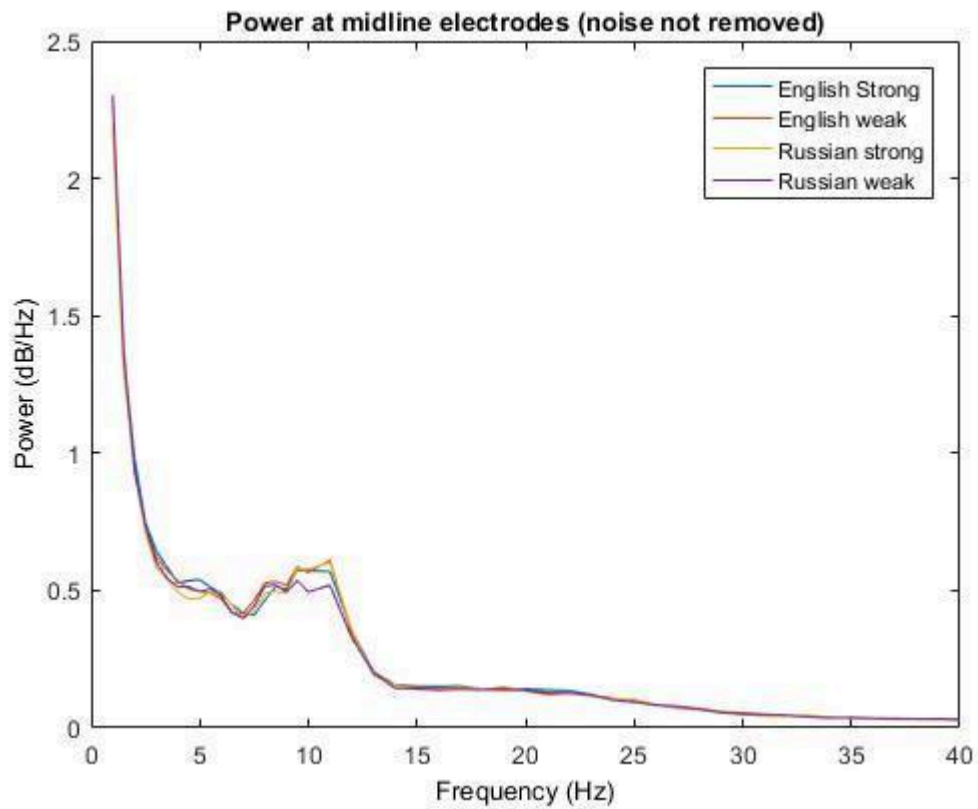


Figure 6. Power spectrum of EEG trials in each condition, averaged over midline electrodes, including 1/f noise. Power (dB/Hz) is plotted as a function of frequency (Hz).

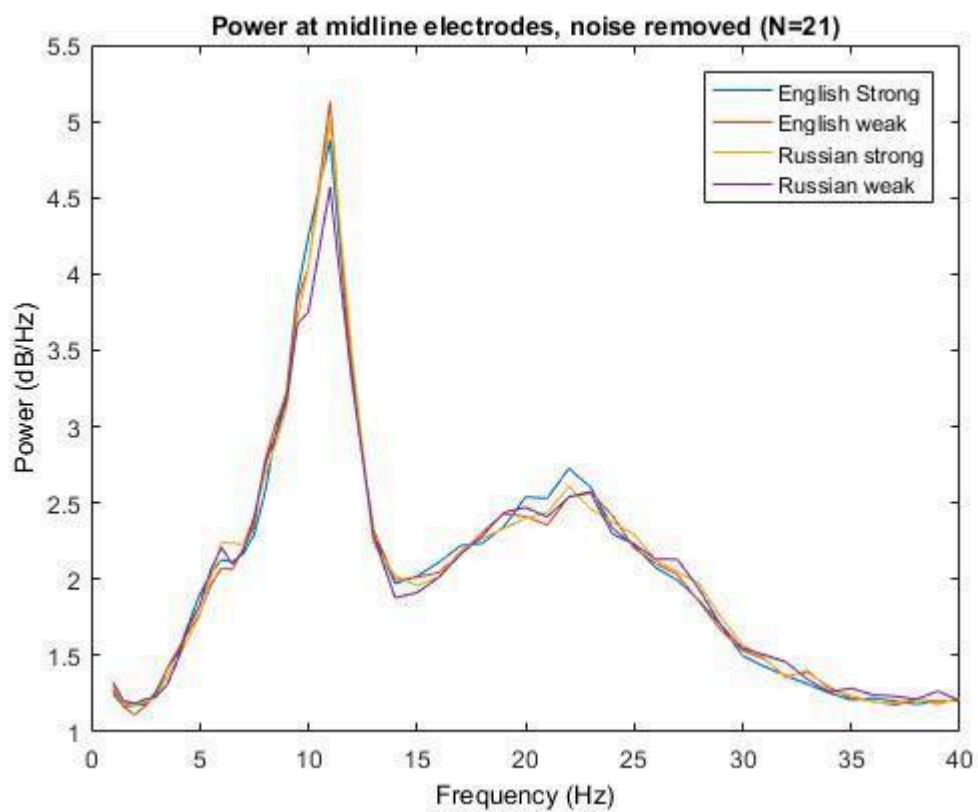


Figure 7. Power spectrum of EEG trials in each condition, averaged over midline electrodes, with 1/f noise removed. Power (dB/Hz) is plotted as a function of frequency (Hz).

To follow from results by Doelling et al (2014), we investigated the phase coherence between the EEG signal and the sound envelope. Unfortunately, this seemed to be quite noisy for all conditions, not showing a peak at any of the frequencies of interest, so we left out any further analyses of significance within specific frequency ranges. Figure 8 depicts an image of the cerebro-acoustic coherence difference between actual trials and randomised ones. There was no difference between the phase coherence to actual stimuli and random ones, neither when conditions were collapsed together, not when these were tested individually.

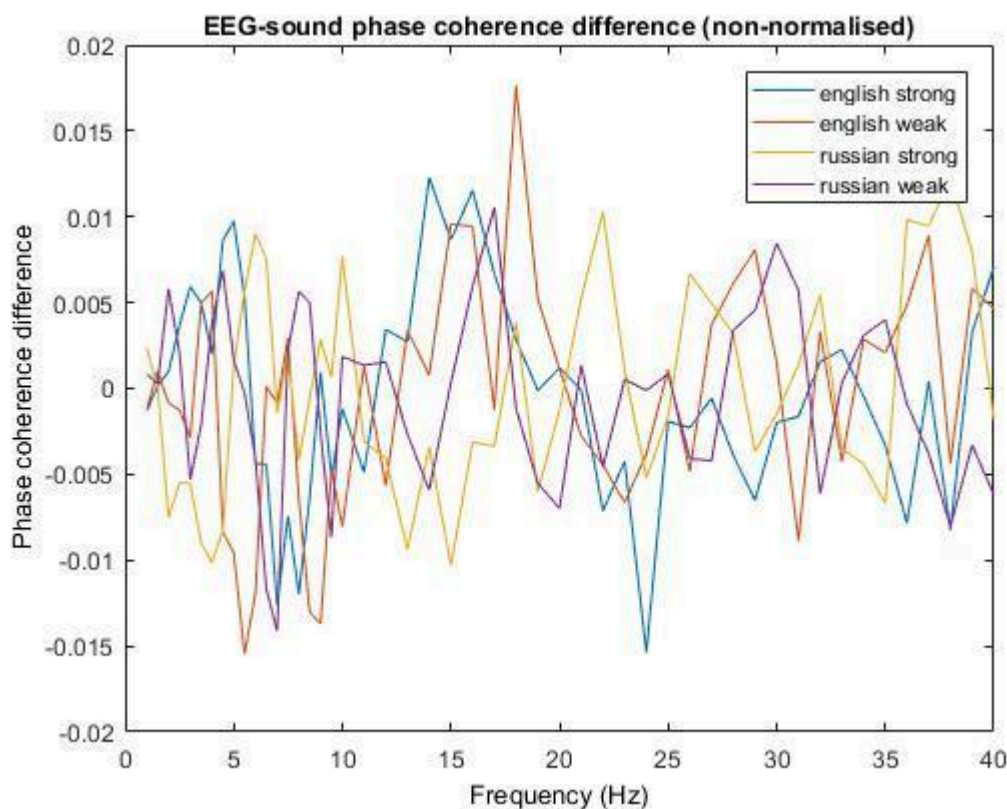


Figure 9 (maybe choose something else). The phase coherence between EEG trials and random stimuli was extracted from that to corresponding sound envelopes. This difference is plotted as a function of frequency, for each condition.

Phase coherence difference also entailed little to no results, **in the sense that the same peaks corresponding to the ones we saw in the power spectrum were noticed but covered by a lot of noise** (e.g., Figure 9). However, when looking at the difference between intra-trial and inter-trial phase coherence, the slight peak in theta is higher for the intra-trial than for the inter-trial phase coherence, even if not significant. A more in-depth investigation showed that in fact, it was the intra-trial coherence in the Strong conditions which was greater than the inter-trial one, and significantly so ($t(20)=2.26$, $p=.017$). This is only true for the theta frequency range, and no significant tests were obtained between or across the other two frequency bands which we investigated. To further investigate the difference between weak and strong stimuli, we obtained the z-scored phase coherence for each of the weak and strong conditions, by subtracting the mean of the inter-trial coherence from that of the intra-trial coherence, and then dividing it by the standard deviation of the

inter-trial coherence. However, this measure did not result in a significant difference between weak and strong stimuli. Figures 10 and 11 show the difference between intra- and inter-trial coherence for strong and weak stimuli, respectively. While the peak for strong stimuli shows a clear peak at about 6 Hz, weak intra-trial coherence seems to be smeared across the entire theta range, so it is possible that the two conditions had different effects on entrainment even if they did not differ between each other.

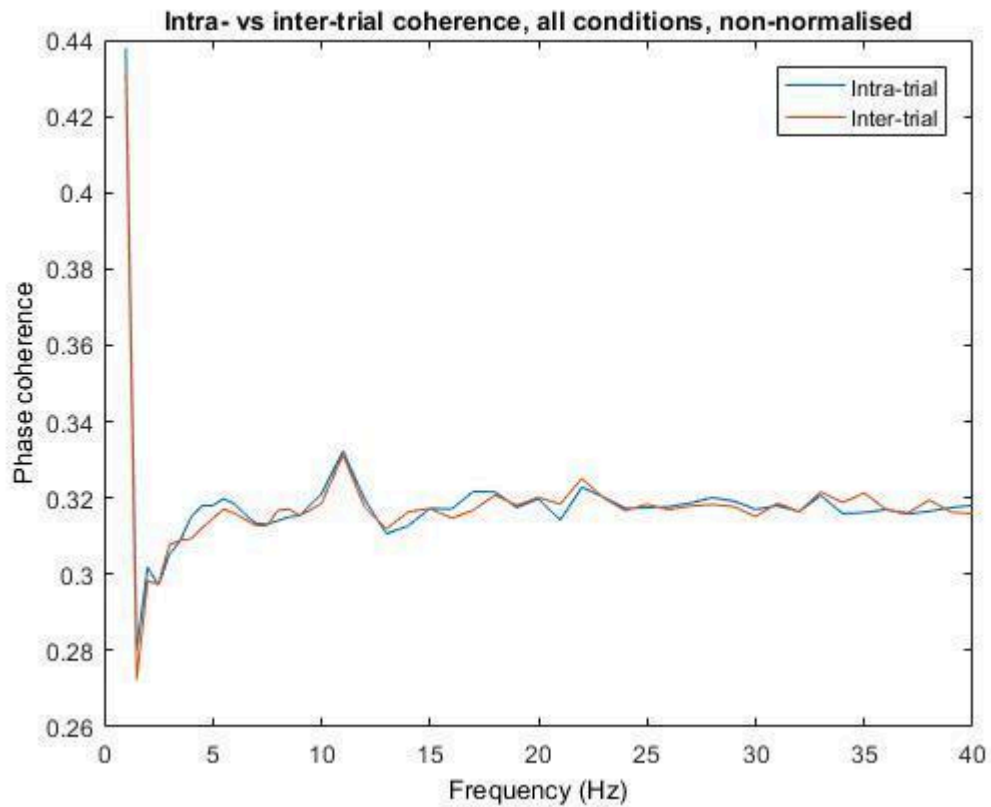


Figure 10. Intra-trial and inter-trial phase coherence are averaged across all conditions and plotted together, at midline electrodes. Phase coherence is plotted as a function of frequency. A peak in alpha can be seen in both cases, but they seem to overlap. A smaller peak in theta is also seen for both, but while slightly greater in the intra-trial phase coherence, this difference is not significant.

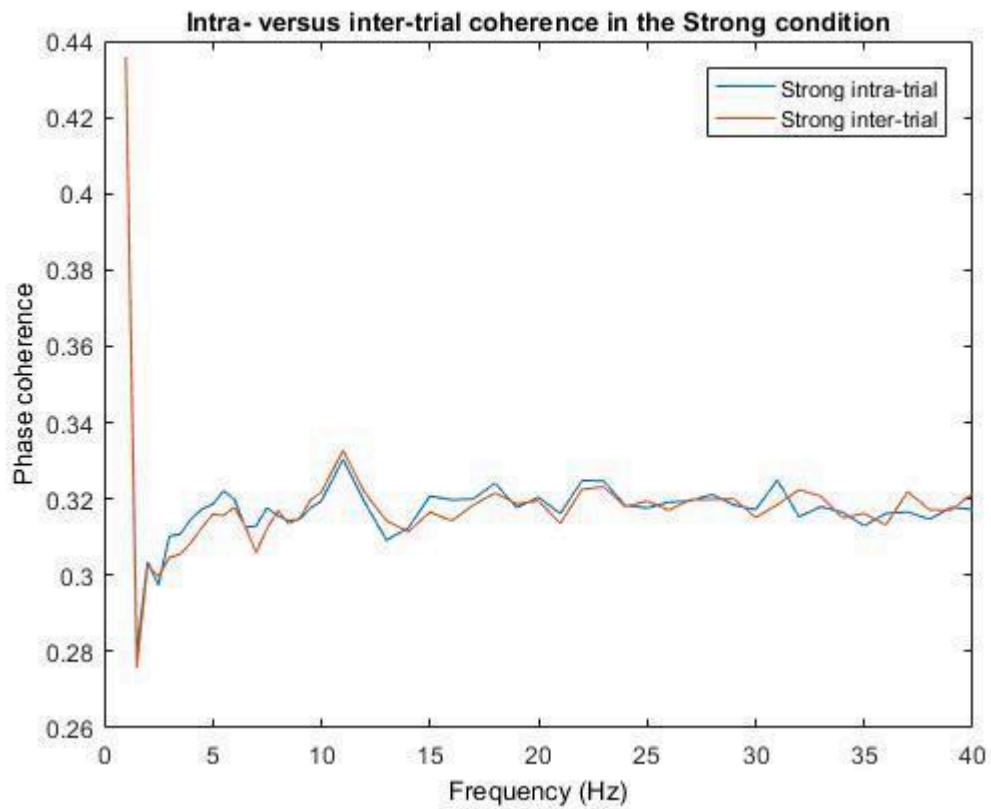


Figure 10. Intra- and inter-trial phase coherence at midline electrodes, averaged across trials to 'strong' stimuli and plotted together.

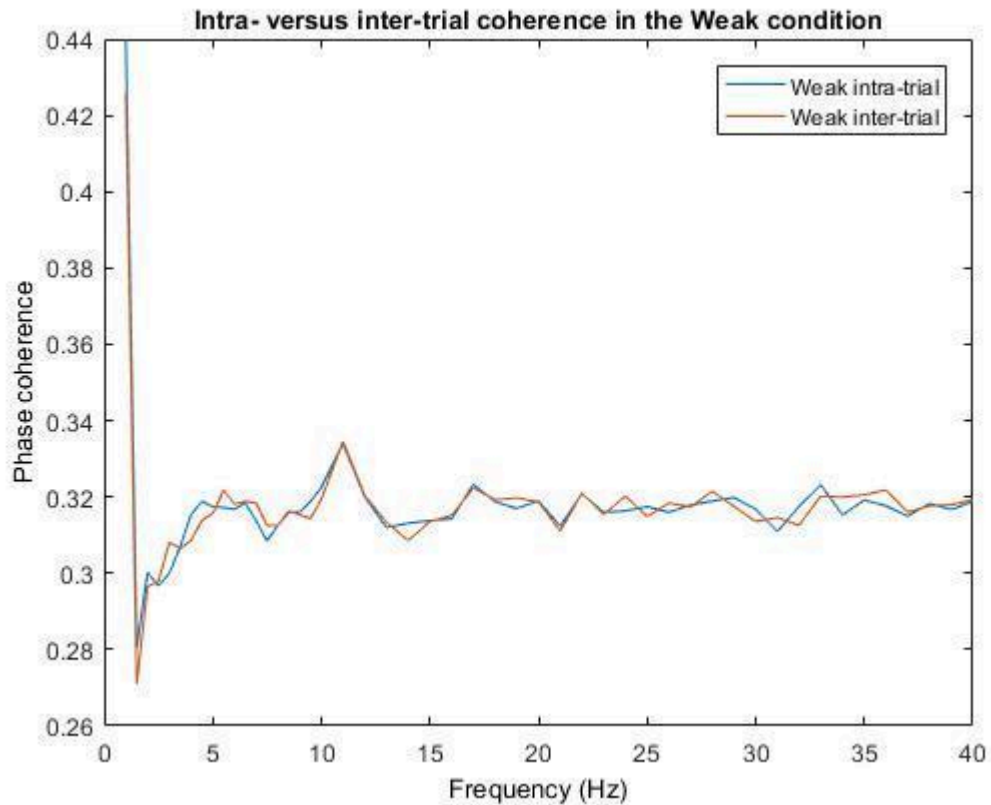


Figure 11. Intra- and inter-trial phase coherence at midline electrodes, averaged across trials to 'weak' stimuli and plotted together.

Similar results were observed in terms of power covariance, in the sense that stronger responses were elicited by intra-trial versus inter-trial power covariance in the theta range only. While these differences may seem small (Figure 12), they are surprisingly significant ($t(20)=2.95, p=.003$). These seem mostly to be driven by results in the Russian condition ($t(20)=3.37, p=.002$) and nearly reached significance in the English condition ($t(20)=1.72, p=.05$) (see Figures 13 and 14). There was a clearer peak in the theta range, in the English condition. However, there was no significant difference between English and Russian either within the intra-trial or the inter-trial ranges. Nonetheless, the z-scored power covariance showed a significant peak at 5.5 Hz, and this is significantly higher than the covariance in the other two frequency ranges, which potentially strengthens the previous results we obtained in this range (see Figure 15).

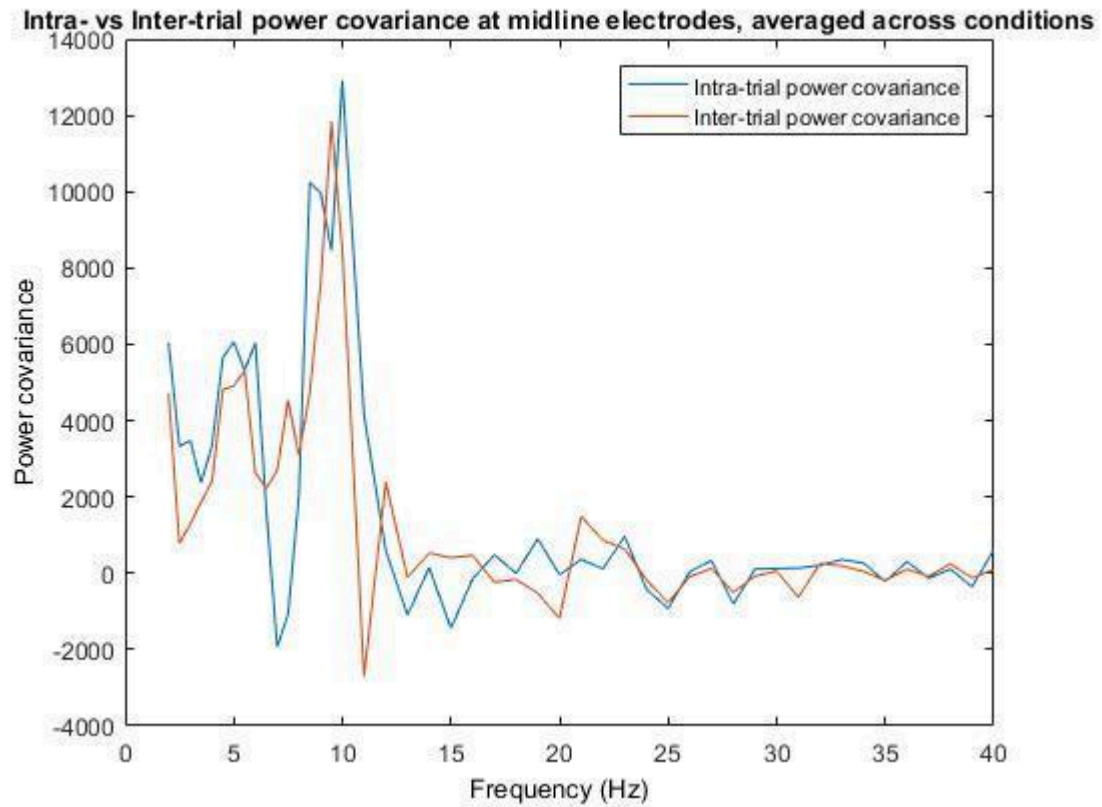


Figure 12. Intra-trial and inter-trial power covariance are plotted together as a function of frequency. Small peaks can be seen in both the theta and gamma frequency ranges, but not gamma. Only the peak in theta is significantly bigger for the intra-trial power covariance compared to the inter-trial one.

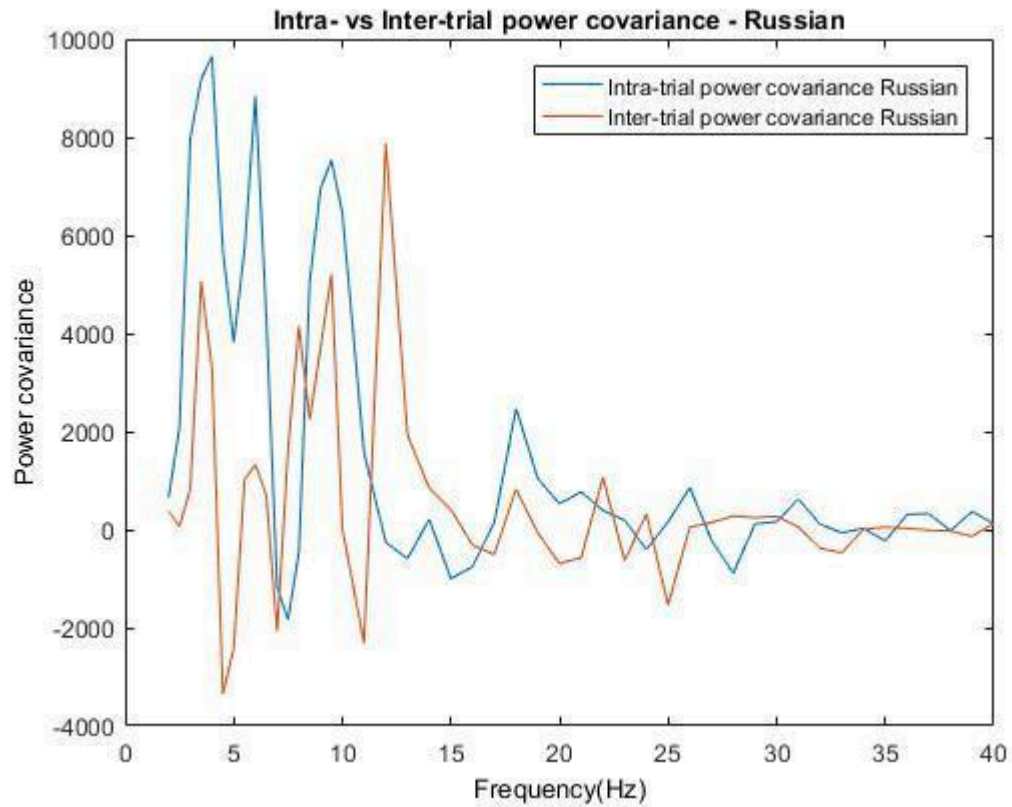


Figure 13. Intra- and inter power covariance at midline electrodes are plotted as a function of frequency, in the averaged Russian condition. Intra-trial covariance is overall greater than inter-trial covariance, but there are two peaks in the 'within' condition.

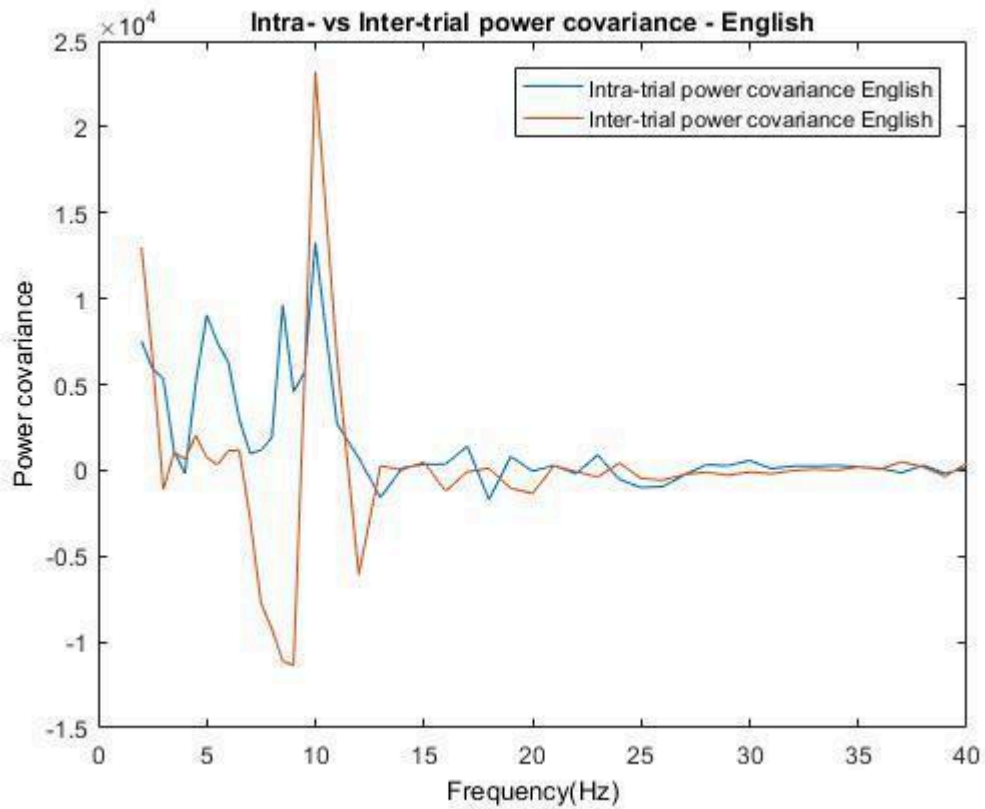


Figure 14. Intra- and inter power covariance at midline electrodes are plotted as a function of frequency, in the averaged English condition. A clear peak can be seen at 5 Hz, for the intra-trial covariance.

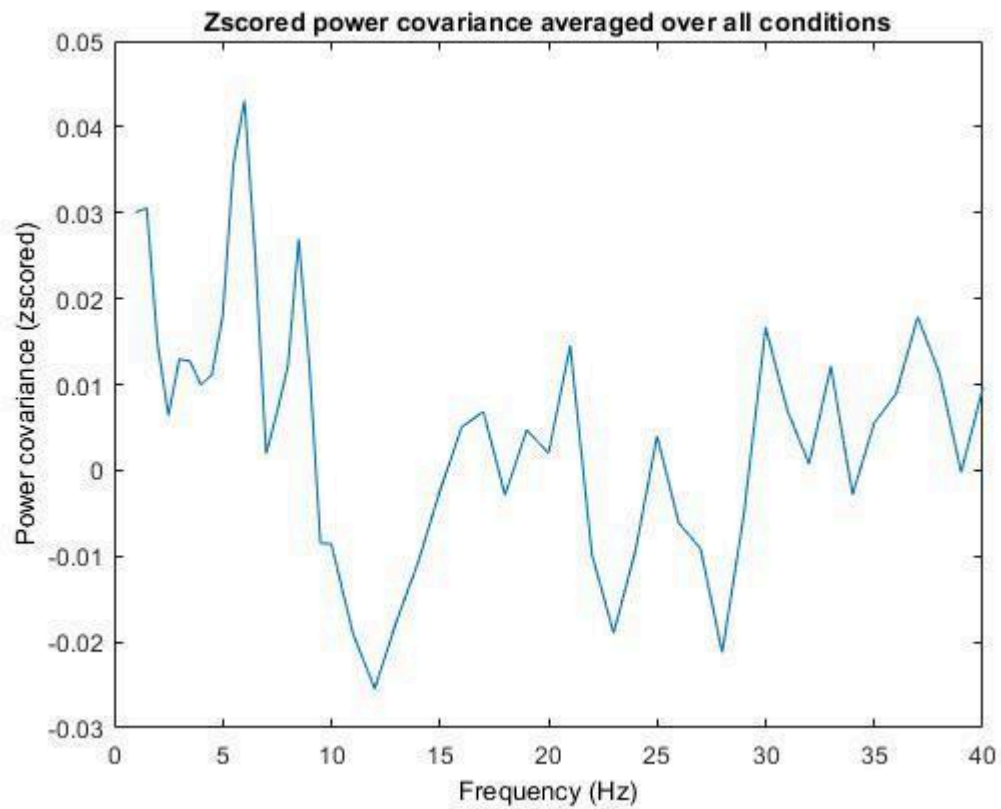


Figure 15. Power dissimilarity function is plotted against frequency (Hz). This is higher in the theta range than in the other two frequency intervals.